

System Prompts

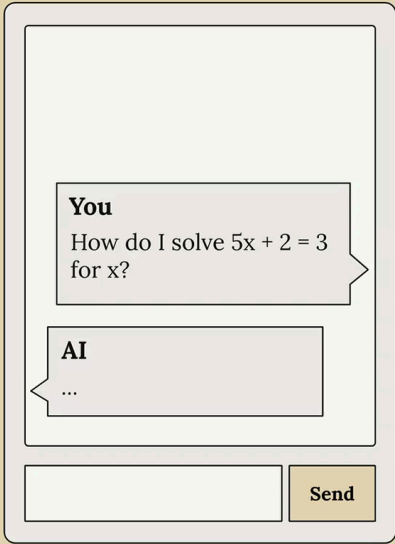
Video Link: [System Prompts](#)

System prompts are a powerful way to customize how Claude responds to user input. Instead of getting generic answers, you can shape Claude's tone, style, and approach to match your specific use case.

Math Tutor Specialist

- **Responses should:**
 - Initially, only give the student hints
 - Patiently walk the student through a solution
 - Show solutions for similar problems
- **Responses should not:**
 - Immediately give a direct answer
 - Tell the student to use a calculator

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Why System Prompts Matter

Consider building a math tutor chatbot. When a student asks "How do I solve $5x + 2 = 3$ for x ?", you want Claude to act like a real tutor, not just spit out the answer. A good math tutor should:

- Initially give hints rather than complete solutions
- Patiently walk students through problems step by step
- Show solutions for similar problems as examples

You definitely don't want Claude to:

- Immediately give direct answers
- Tell students to just use a calculator

How System Prompts Work

System prompts

Customizes Claude's tone and style of response

System prompts provide Claude guidance on how to respond

Claude will try to respond in the same way someone in the specified role would respond

Helps keep Claude on task

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```
system_prompt=""
You are a patient math tutor. Do not directly
answer a student's questions. Guide them to a
solution step by step.
""

client.messages.create(
    model=model,
    messages=messages,
    max_tokens=1000,
    system=system_prompt
)
```

System prompts provide Claude with guidance on how to respond. You define them as plain strings and pass them into the create function call. The key benefits are:

- System prompts provide Claude guidance on how to respond
- Claude will try to respond in the same way someone in the specified role would respond
- Helps keep Claude on task

Here's the basic structure:

```
system_prompt = """
You are a patient math tutor.
Do not directly answer a student's questions.
Guide them to a solution step by step.
"""

client.messages.create(
    model=model,
    messages=messages,
    max_tokens=1000,
    system=system_prompt
)
```

Seeing the Difference

Without a system prompt, Claude gives a complete step-by-step solution immediately. This might be helpful, but it doesn't encourage the student to think through the problem themselves.

With the math tutor system prompt, Claude's response changes dramatically. Instead of providing the full solution, Claude asks guiding questions like "What do you think would be a good first step to isolate x?"

Consider what operation we might need to perform on both sides to start moving terms around."

Building a Flexible Chat Function

Rather than hard-coding system prompts, you can make your chat function more reusable by accepting system prompts as parameters:

```
def chat(messages, system=None):
    params = {
        "model": model,
        "max_tokens": 1000,
        "messages": messages,
    }

    if system:
        params["system"] = system

    message = client.messages.create(**params)
    return message.content[0].text
```

This approach handles an important detail: Claude's API doesn't accept `system=None`, so you need to conditionally include the system parameter only when it's provided.

Now you can call your chat function with or without a system prompt:

```
# Without system prompt
answer = chat(messages)

# With system prompt
system = """
You are a patient math tutor.
Do not directly answer a student's questions.
Guide them to a solution step by step.
"""
answer = chat(messages, system=system)
```

System prompts are essential for creating AI applications that behave consistently and appropriately for their intended purpose. They transform generic AI responses into specialized, role-appropriate interactions.